Coastal Erosion Study in Canterbury Region

Data601 – 25X

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Abstract

Storm wave conditions have a significant impact on coastal erosion along the Canterbury coast, but measuring this relationship is still challenging because of sparse observations and uncertainty in offshore wave models. This study combines satellite-derived shoreline positions, hindcast wave model output, and buoy observations to examine how various storm types affect shoreline change. Based on wave height, duration, and direction, storm events were initially recognized and categorized into physically significant storm types. In order to measure biases in storm intensity, duration, and approach direction, the hindcast model was then verified using buoy observations. In order to evaluate how various storm types affect erosion, these verified storm metrics were then connected to monthly shoreline change obtained from CoastSat imagery. Lastly, statistical models were applied to forecast the occurrence of storm-related erosion. The findings indicate that storms can be classified into three different types with varying intensities and durations, and that these types show systematically different responses along the shoreline. Storm direction and duration showed reasonable agreement, but hindcast wave heights were biased high in comparison to buoy observations. Despite significant natural variability, erosion models showed limited but non-negligible predictive skill, suggesting that storm characteristics carry quantifiable information about shoreline change. This work demonstrates the potential and limitations of storm-based erosion prediction and offers a repeatable framework for connecting offshore storm forcing to coastal erosion.

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1. Introduction

1.1 Background and Motivation

Coastal erosion is a global concern due to its significant result of climate change and changes in the natural fluctuations. Coastal erosion can also pose major risks to coastal infrastructure, environmental reserves, wildlife and human communities. In Aotearoa New Zealand, coastal erosion has become a serious problem on high energy coasts (Canterbury Region). The factors causing coastal erosion are multifaceted and include hydroelectric dams which have reduced sediment transport capacity from rivers to the sea, coastal sediment budgets being disrupted by mining and the removal of sand and long term increases in sea level creating an ongoing risk. Storms are however the primary cause of short-term damage. Therefore, it is necessary to understand the physical processes linking atmospheric forcing, ocean waves and shorelines to improve our ability to assess and manage coastal hazards.

The combination of advancements in both observationally collected data and model data has provided researchers with a greater number of opportunities to study coastal erosion process than were previously available. The satellite derived shoreline dataset is providing the ability to monitor shoreline position over extended time frames. Wave buoys and numerical wave hindcast are providing a wealth of detail on offshore and nearshore wave condition. Additionally, atmospheric reanalysis and high-resolution weather models are being used to identify the characteristics of storms that produce extreme wave events. A major challenge in integrating these datasets is their differing spatial and temporal resolutions and data formats.

1.2 Research Aims and Objectives

The purpose of this research is to gain a better understanding of how waves interact with the atmosphere and how both can cause coastal erosion by combining data from three different sources: the shoreline, wave action and atmospheric conditions. This research addresses three main objectives:

- Firstly, storm events will be identified and classified based on their atmospheric and wave characteristics.
- Secondly, the performance of wave models will be assessed through comparison with buoy observations
- Lastly the relationships between storm types and observed shoreline change will be explored.

The goal for this research is to help produce improved forecasts of coastal hazards in New Zealand and to provide information to aid coastal management.

1.3 Research Questions

Based on these objectives, this study addresses the following research questions:

1. Can storm events be objectively identified and classified using atmospheric and wave characteristics?
2. How accurately does the regional wave hindcast reproduce observed storm conditions?
3. Do different storm types correspond to systematic differences in observed shoreline change?

2. Method

2.1 Workflow Overview

This study follows a structured workflow designed to link offshore storm characteristics to observed shoreline change along the Canterbury coast. The analysis integrates storm detection, storm classification, model validation, shoreline response analysis, and erosion prediction within a single, coherent framework.

First, storm events were identified independently from both ECan wave buoy observations and a regional New Zealand wave hindcast using a peak-over-threshold approach applied to significant wave height. Each storm event was summarised using physically meaningful metrics, including duration, peak wave height, and mean wave direction.

Second, identified storms were grouped into storm types based on their wave characteristics to derive a storm typology that distinguishes between background conditions and more energetic storm regimes. This typology provides a physically interpretable basis for analysing differences in coastal response.

Third, storm characteristics derived from the wave hindcast were validated against ECan buoy observations to quantify model bias and uncertainty in storm intensity, duration, and direction. This validation step ensures that limitations in the hindcast representation of storms are explicitly accounted for in subsequent analyses.

Fourth, shoreline change was quantified using satellite-derived shoreline positions obtained from the CoastSat framework. Monthly median shoreline positions were calculated along multiple transects and converted into monthly shoreline change values, allowing shoreline response to be analysed at timescales consistent with storm forcing.

Finally, storm metrics and shoreline change were integrated into a single dataset to assess whether storm characteristics can be used to predict erosion. A logistic regression model was applied to estimate the probability of erosion occurrence, enabling storm–erosion relationships to be interpreted in a probabilistic rather than deterministic context.

2.2 The ECan Wave Buoy, Storm Detection and Clustering Methodology

The ECan (Environment Canterbury) wave buoy is a single-point measurement moored in about 76 meters of water, 17 km east of Le Bons Bay on Banks Peninsula. Its coordinates are approximately 43°45' S, 173° E, focusing on local nearshore wave data in the Canterbury coastal region. It provides hourly wave measurements from 6 February 1999 to 30 September 2025. The dataset consists of quality-controlled spectral wave buoy observations, including significant wave height (Hs), peak wave period (Tp), mean wave direction, and spectral peakedness. Date and time variables were combined into a single datetime field to enable time-series analysis.

The Preprocessing steps that were carried out on the dataset include removing values that were physically implausible, handling of missing values and sorting the data chronologically.

Storms were identified using the Peak Over Threshold (POT) method, which aims to identify storm events in a continuous wave record that exceed a significant wave height threshold; that are maintained for a minimum storm duration; and that are separated by a minimum storm recurrence interval. To capture frequent moderate storms that cause cumulative erosion, a minimum storm duration of 24 hours was used. According to the regional progression of synoptic events a 24-hour recurrence interval was used. For each identified storm, summary characteristics were derived including duration, maximum and mean significant wave height, mean peak period, mean wave direction, and mean spectral peak.

Once storm events were separated out the timeseries was normalized in preparation for clustering using K means. K means is an unsupervised learning method that partitions a dataset into k predefined clusters effectively grouping similar data points around cluster centroids. It does this by minimizing the within-cluster sum of squares (inertia). The optimal k to be used for the clustering was determined using the elbow method. The elbow method involves plotting the within-cluster sum of squares (WCSS) against various K values and identifying the "elbow" point where adding more clusters yields diminishing returns. The elbow point was when k =3. Thus, the number of clusters considered in the k means analysis was 3. This resulted in 3 distinct storm types -Cluster 0, Cluster 1 and Cluster 2.

With relatively lower wave heights and shorter dominant periods, Cluster 0, also known as Lower-Energy Short-Period Storms, is concentrated close to the lower bound of the storm threshold. These events meet the storm definition even though they are less energetic than the other clusters; they are sustained but lower-energy storms.

Cluster 1, known as Moderate-Energy Mixed Storms, has a wide range of peak periods and an intermediate range of wave heights. This cluster, which reflects mixed wave conditions influenced by both wind-sea and swell components, is the most prevalent type of storm.

Cluster 2, also known as High-Energy Swell Storms, is distinguished by waves that are consistently higher throughout moderate to long peak periods, typically surpassing 4 m. These storms are a sign of well-developed swell-dominated systems and are the most energetic wave conditions.

While alternative clustering approaches such as density-based methods or self-organising maps have been applied in wave-climate studies, a simple clustering framework was adopted here to prioritise physical interpretability and direct comparison with shoreline response. The aim of the storm typology was not to optimise cluster separation mathematically, but to derive physically meaningful storm regimes suitable for subsequent coastal impact analysis.

2.3 Hindcast–Buoy Validation

To accurately model erosion and assess coastal hazard it is important to have an accurate representation of wave conditions. Numerical wave models are used to reconstruct historical and present-day wave climates in regions where long-term observational data are limited.

The New Zealand (NZ) wave Hindcast is one such model. It covers the period from 1 January 1993 to 31 December 2019. It provides gridded estimates of wave parameters around New Zealand's coastline. It does this at consistent temporal intervals. To assess the reliability of the model in representing storm conditions, storm characteristics derived from the model were validated against in situ observations from the ECan wave buoy. Four variables from the model were involved in this analysis. These variables are significant wave height (Hsig), peak wave period (RTpeak), mean wave direction (Dir), and directional spread (FSpr).

To ensure that a like-for-like comparison between both datasets, they were harmonized in space and time. Time stamps were converted to Coordinated Universal Time (UTC) and then aggregated to an hourly resolution. The hindcast grid point closest to the ECan buoy location were chosen to represent the simulated wave conditions at the buoy site.

A paired dataset with each row representing the same hour in time for both the buoy and hindcast was created by merging the observed and modelled datasets using exact hourly timestamp matching. The only variables kept were those that were physically comparable By using this method, variations between the datasets are guaranteed to represent model performance rather than temporal or spatial discrepancies.

To ensure independence from the hindcast model, storm events were identified solely from buoy observations. The observed storm period were then used to define the exact time windows used to extract and analyse corresponding data from the model. The definition of a storm remained the same as was used when identifying storms in the ECan buoy dataset.

Individual storm events identified in the hindcast were temporally matched with corresponding storm events detected in the buoy record. For each matched storm pair, differences between hindcast and observed storm metrics were evaluated to quantify systematic biases and variability in model performance.

Validation was undertaken to identify whether the hindcast accurately represents not only the occurrence of storms, but also their physical characteristics. Particular attention was given to storm duration and wave height, as biases in storm persistence or intensity can lead to over- or underestimation of cumulative wave forcing acting on the coast. Storm wave direction was also assessed, as directional accuracy is critical for understanding alongshore sediment transport and shoreline response.

The results of this validation provide essential context for interpreting storm typology and erosion analyses. Any discrepancies between hindcast and buoy-derived storm characteristics directly influence storm classification and subsequent erosion modelling, and therefore must be explicitly considered when drawing conclusions about storm–shoreline relationships.

2.4 Shoreline Change Analysis

To measure how the coast moved over time, we used monthly shoreline position estimates derived from satellite imagery (CoastSat). These shoreline positions were extracted along a set of fixed coastal transects. For each transect, shoreline position is represented as a distance along the transect, and month-to-month differences were computed to quantify shoreline movement. We define monthly shoreline change as the difference in shoreline position from one month to the next, expressed in metres (m) where negative values =

erosion (shoreline moved landward) & Positive values = accretion (shoreline moved seaward)

Because multiple transects were used, the study summarised the coastline-wide response using a single monthly value: mean_shoreline_change = the average shoreline change across all available transects for that month. This gives one interpretable number per month that can be directly paired with monthly storm summaries.

Storms occur at sub-daily scales, while satellite shoreline observations are most reliable at monthly resolution. To make datasets comparable, all storm information was aggregated to the same monthly time base as the shoreline record. For each month, storm conditions were summarised using storm metrics such as: storm duration (e.g., ecan_storm_hours), storm intensity (e.g., ecan_max_Hs, ecan_mean_Hs), storm direction (e.g., ecan_mean_dir)

This monthly aggregation produces a modelling-ready dataset where each row corresponds to one month containing: shoreline change (mean_shoreline_change), and storm characteristics for that month.

After storms were clustered into three storm types (storm typology), we tested whether shoreline behaviour differed across types by grouping monthly shoreline change by storm_type.

This was examined using: summary statistics by storm type (count, mean, median, spread), and a boxplot of mean_shoreline_change by storm type to visualise:

- the typical shoreline response for each storm regime,
- how variable each response is, and
- whether extreme erosion months are concentrated in specific storm types.

This analysis provides an intuitive answer to a coastal-management question:
which storm regimes are most associated with high erosion risk?

2.5 Erosion Prediction Modelling

The prediction task was framed as a simple, practical question: Given the storm conditions in a month, can we estimate whether that month is likely to be erosive? Instead of predicting the exact shoreline change magnitude (which is strongly influenced by many unobserved coastal factors), we performed binary classification: erosion vs not erosion. A binary indicator variable was used:

- erosion_month = 1 if mean_shoreline_change is negative (net erosion that month)
- erosion_month = 0 if mean_shoreline_change is zero or positive (stable/accretion)

This definition matches the coastal interpretation: we care about whether erosion occurred, not only how large it was.

The model used storm descriptors that represent physically meaningful drivers of wave impact:

- Intensity: wave height measures (mean and maximum significant wave height)
- Persistence: total storm duration in hours
- Direction: mean storm approach direction

Because direction is circular (0° and 360° are the same), direction was represented using trigonometric components: dir_sin = sin(direction) & dir_cos = cos(direction). This avoids artificial “breaks” at 0/360 and allows direction to contribute smoothly in the model.

We used logistic regression because it is: appropriate for a binary outcome (erosion vs not), interpretable (shows which predictors increase or decrease erosion risk) and easy to communicate to a non-technical audience as a probability-of-erosion model. In plain terms, the model estimates: Probability (month is erosive) based on storm characteristics.

To test performance fairly, data were separated into training and testing sets. The model was fitted on the training portion and evaluated on unseen months. Performance was assessed using a ROC curve and its AUC (Area Under the Curve), because:

- AUC summarises discrimination skill across all probability thresholds.
- AUC = 0.50 means random guessing.
- AUC > 0.50 means the model provides useful signal.

Even with accurate storm metrics, shoreline response remains variable because erosion depends on more than offshore wave forcing, including:

- pre-storm beach condition (antecedent state),
- sediment availability and transport,
- storm sequencing (clusters of storms),
- nearshore processes not fully captured offshore.

This is why we interpret the model as probabilistic risk estimation, not deterministic prediction.

3. Data

This study integrates wave, atmospheric, and shoreline datasets to examine storm-driven coastal erosion along the Canterbury coast. Three primary data sources were used: ECan wave buoy observations, a regional New Zealand wave hindcast, and satellite-derived shoreline positions from the CoastSat framework.

3.1 Wave Buoy Observations

Offshore wave observations were obtained from the ECan wave buoy located off the Canterbury coast. The buoy provides in situ measurements of significant wave height, wave period, and wave direction at high temporal resolution. These observations represent the most direct measurements of storm wave conditions and were used as the reference dataset for characterising observed storm events and for validating hindcast model output.

Prior to analysis, buoy time series were inspected for missing or invalid values. Periods containing data gaps or quality-control flags were excluded to ensure that storm detection was based on reliable observations. All wave variables were retained in consistent units and temporal resolution for subsequent storm-level summarisation.

3.2 Wave Hindcast Data

Wave hindcast data were obtained from a regional numerical wave model covering the Canterbury shelf. The hindcast provides continuous spatial and temporal coverage of offshore wave conditions, including significant wave height, wave period, and mean wave direction. Hindcast data were extracted at the grid point nearest to the buoy location to facilitate direct comparison with observed storm characteristics.

The hindcast time series was screened for missing values and aligned temporally with the buoy record. Storm-level metrics were calculated from the hindcast using the same procedures applied to the buoy data to ensure consistency between observed and modelled storm representations.

3.3 Shoreline Position Data

Shoreline position data were derived using the CoastSat framework, which applies automated shoreline detection to optical satellite imagery. The national CoastSat site repository was first filtered to retain only sites located within the Canterbury region.

For each retained site, shoreline positions were extracted along predefined transects. Monthly median shoreline positions were calculated to reduce the influence of tidal variability, cloud contamination, and short-term noise. Transects with insufficient satellite coverage were excluded to ensure robust shoreline change estimates. Monthly shoreline change was then computed as the difference between successive monthly median positions and expressed in metres.

3.4 Data Integration and Aggregation

All datasets were temporally aligned to a common monthly time base to enable comparison between storm forcing and shoreline response. Storm metrics derived from both buoy and hindcast data were aggregated to monthly summaries. When multiple storm events occurred within a single month, storm characteristics were combined to represent overall monthly storm forcing.

The resulting cleaned and merged dataset links storm characteristics, model validation metrics, and shoreline change observations. This integrated dataset forms the basis for storm typology development, hindcast–buoy validation, and erosion prediction analyses presented in subsequent sections.

4. Results

4.1 Storm typology

Storm types were defined using ECan wave buoy data.

Figure 1: K-Means Storm Clusters Showing Wave Height–Period Characteristics

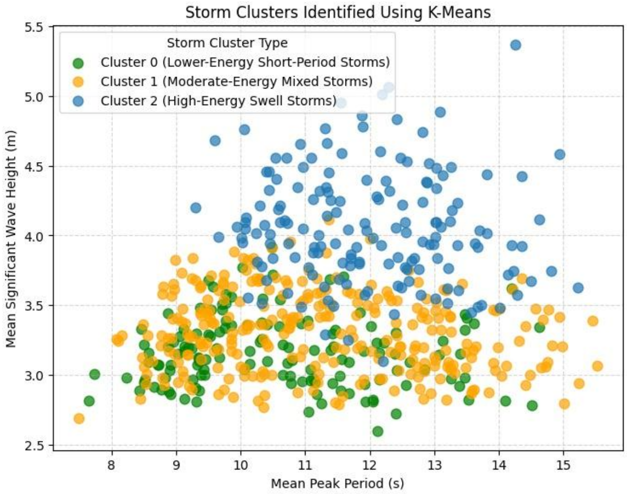

Figure 1 shows storm events classified using K-means clustering based on their mean significant wave height and mean peak period. Each dot is an individual storm that satisfies the storm definition criteria, colours are used to indicate the three identified storm clusters.

Cluster 0 labelled as Lower-Energy Short-Period Storms is concentrated near the lower wave-height threshold with generally shorter peak periods, representing less energetic but still sustained storm events.

Cluster 1, labelled as Moderate-Energy Mixed Storms, has a wide range of peak periods and an intermediate range of wave heights. This cluster, which reflects mixed wave conditions influenced by both wind-sea and swell components, is the most prevalent type of storm.

Cluster 2, labelled as High-Energy Swell Storms, is distinguished by waves that are consistently higher throughout moderate to long peak periods, typically surpassing 4 m. These storms are a sign of well-developed swell-dominated systems and are the most energetic wave conditions.

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Figure 2. K-Means Storm Clusters in Duration–Wave Height Space.

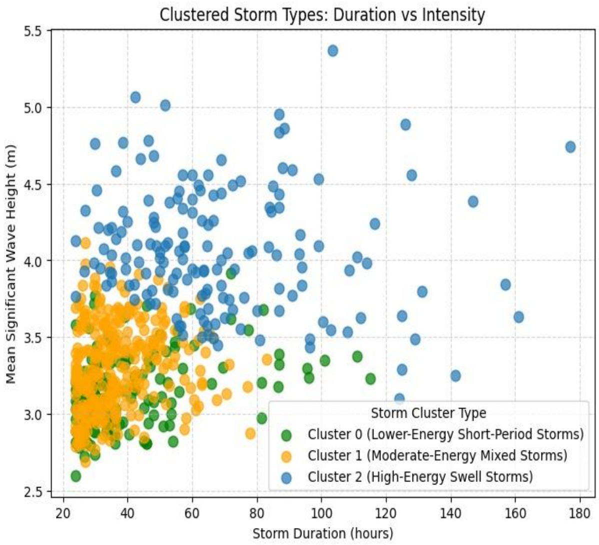


Figure 2 illustrates the relationship between storm duration and mean significant wave height for the three storm clusters identified using K-means clustering. Each dot represents an individual storm event that meets the storm definition criteria, with colours indicating the assigned storm cluster.

Wave heights and durations are typically lower in Cluster 0 (Lower-Energy Short-Period Storms), which is frequently near the minimum storm duration threshold. Even though these occurrences last long enough to qualify as storms, their diminished duration and intensity suggest a relatively smaller impact on coastal erosion.

Cluster 1 (Moderate-Energy Mixed Storms) is characterised by moderate wave heights and mainly short to moderate durations, most commonly between 24 and 60 hours. This cluster represents typical storm conditions that contribute to ongoing background coastal processes rather than extreme episodic erosion.

While continuously maintaining higher significant wave heights, Cluster 2 (High-Energy Swell Storms) displays the broadest range of storm durations, including multiple long-lasting events lasting more than 100 hours. These storms exhibit a combination of high intensity and long duration, indicating a higher likelihood of coastal impact as a result of persistent wave forcing.

Storms with comparable durations can have noticeably different wave heights, even though there is some overlap in duration between clusters. This highlights the need to use more than one variable when characterizing storm impacts and reinforces the importance of considering the persistence of a storm as measured by its duration and the intensity of the storm when assessing storm-driven coastal processes.

Figure 3. K-Means Storm Clusters in Direction–Wave Height Space

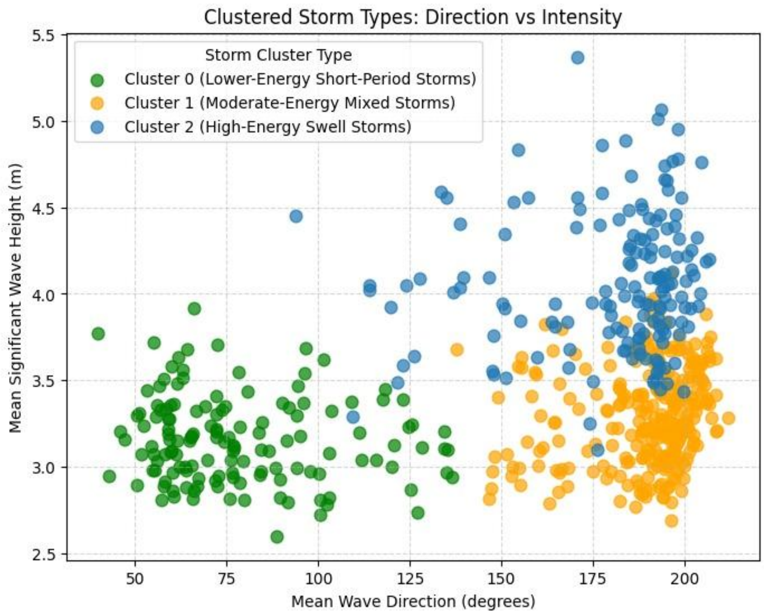


Figure 3 shows the relationship between mean wave direction and mean significant wave height for the storm events classified using K-means clustering. Each dot represents an individual storm that satisfies the storm definition criteria, with colours indicating the assigned storm cluster.

Cluster 0 (Lower-Energy Short-Period Storms) is mainly associated with southerly to south-westerly wave directions and exhibits the highest significant wave heights. It represents energetic swell-dominated storm conditions and is likely to be associated with increased coastal impact.

Cluster 1 (Moderate-Energy Mixed Storms) is characterized by lower wave heights and is likewise primarily concentrated within southerly directional sectors. These storms exhibit mixed wave characteristics, which are more typical storm conditions.

With consistently lower wave heights, Cluster 2 (High-Energy Swell Storms) mostly occurs under easterly to east-southeasterly wave directions. In comparison to southerly storm types, these events are less energetic and are probably going to have less erosive influence, even though they still meet the storm definition criteria.

The figure shows a clear correlation between storm intensity and direction, despite some overlap in wave direction between clusters. The majority of the most intense storm events originate from southerly sectors, underscoring the significance of wave direction as a critical component of storm-driven coastal processes.

4.2 Hindcast versus buoy storm validation

Our storm types are defined using the hindcast wave model, but the real ocean conditions at the Canterbury coast are best represented by the ECan wave buoy. For that reason, we first check how well the hindcast reproduces the same storms seen by the buoy, in terms of wave height, storm duration, and wave direction. This validation matters because any bias in the hindcast will flow into the storm typology and the later storm–erosion results.

To perform this validation, individual storm events identified in the hindcast were matched in time to storms detected in the ECan buoy record. For each matched storm pair, differences between hindcast and buoy values were calculated for wave height, duration, and mean direction. Bias represents the average difference (hindcast minus buoy), RMSE represents the typical size of the error, and correlation measures how well the hindcast captures storm-to-storm variability. Here, the “model” refers to the numerical wave hindcast output itself, rather than a statistical or machine-learning model.

Table 1. Validation statistics comparing hindcast and ECan buoy storm properties (Hs, storm hours, mean direction).

	Bias	RMSE	Correlation
Hs	2.025641	2.075863	-0.01596
Storm hours	-13.1525	20.66576	0.135908
Direction	28.83307	46.68094	0.338918

Table 1 summarises the level of agreement between the hindcast and ECan buoy storm characteristics using three complementary statistics: bias, RMSE, and correlation. Together, these metrics describe not only whether the hindcast systematically differs from observations, but also how large those differences are and whether the hindcast captures relative storm-to-storm variability. Three key patterns emerge from this comparison.

First, significant wave height is strongly overestimated by the hindcast. The large positive bias (approximately +2.03 m) indicates that, on average, the hindcast represents storms as substantially more energetic than those recorded by the buoy. This is a critical result, because wave height is a primary driver of coastal energy input.

Figure 4 clearly illustrates this bias. Most points lie well above the 1:1 reference line, showing that for the same storm events, buoy-observed wave heights are typically in the range of 2.5–3.0 m, while the hindcast frequently predicts values between 4 and 5 m. This systematic offset implies that if hindcast wave heights were used directly to represent coastal forcing, storm energy would be consistently overestimated. As a result, any erosion analysis based solely on raw hindcast wave height would exaggerate the physical impact of storms on the coastline.

Second, storm duration is compressed in the hindcast relative to buoy observations. The negative bias in storm hours (approximately −13.15 hours) indicates that storms tend to be represented as shorter-lived in the hindcast.

Figure 4. Hindcast versus buoy storm mean significant wave height.

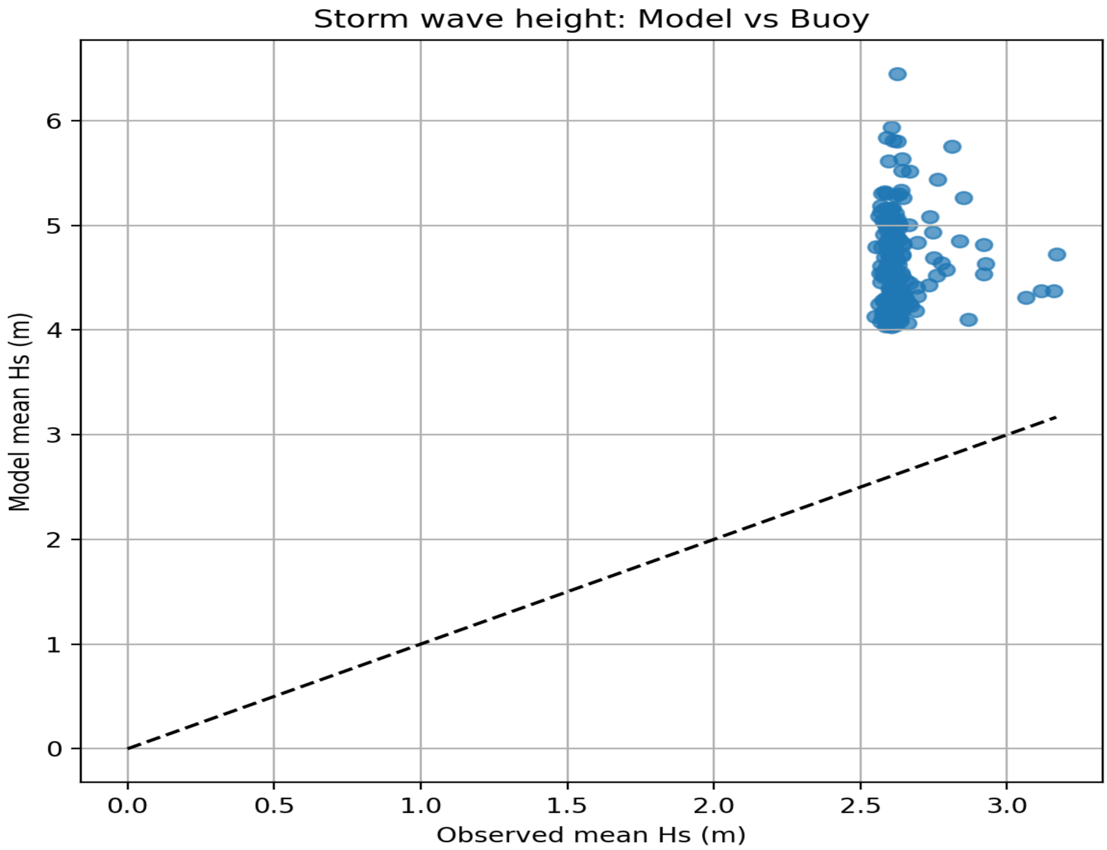


Figure 5. Hindcast versus buoy storm duration.

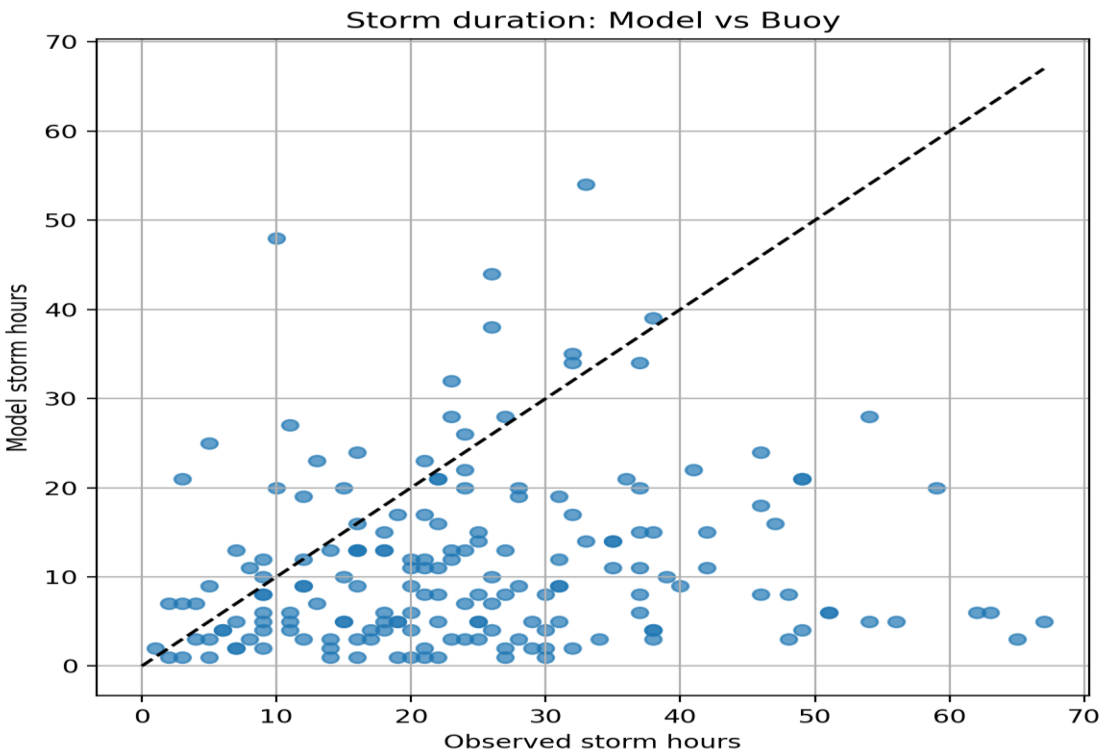
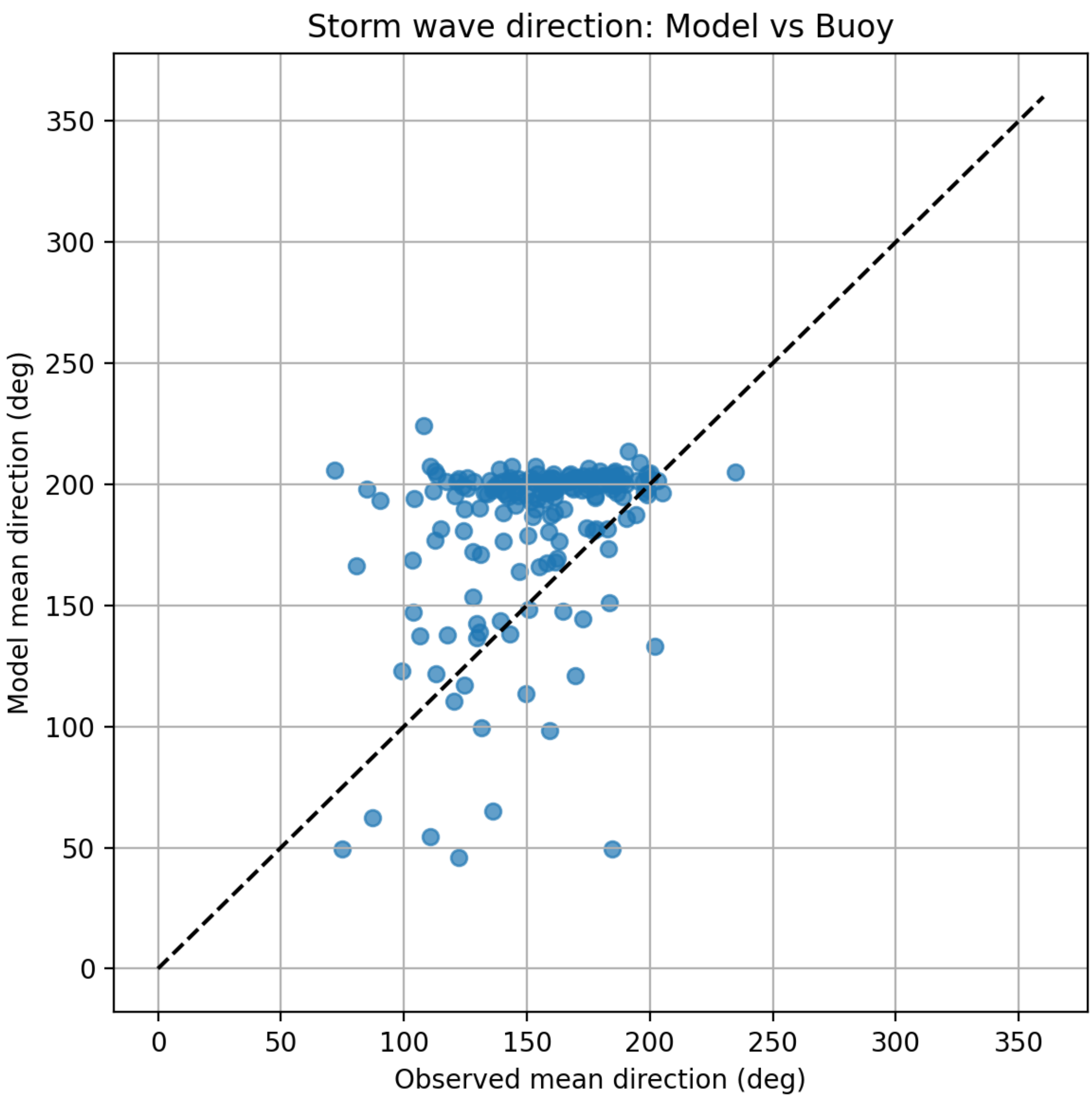


Figure 5 shows that buoy-observed storms span a wide range of durations, from short events lasting only a few hours to prolonged storms exceeding 60 hours. In contrast, hindcast storm durations cluster much more tightly, typically between 5 and 20 hours. This suggests that the hindcast smooths or truncates longer storm events, failing to fully capture storm persistence. Because storm duration influences cumulative wave energy delivered to the coast, this limitation affects the ability of the hindcast to represent prolonged erosive forcing accurately.

Third, storm wave direction is reproduced more reliably, but with reduced variability.

Figure 6. Hindcast versus buoy storm mean wave direction.



As illustrated in Figure 6, the primary directional source of both the buoy and hindcast storm data has been from the southerly to south-westerly directions; therefore, we conclude that the hindcast was able to successfully simulate the predominant storm direction. However, there is a smaller directional spread for the hindcast versus the buoy observations. The reduced variability indicates that while both datasets have similar storm approach direction, the hindcast does not adequately simulate day-to-day differences (i.e., between events) in storm approach angles; these differences in angle can be important for determining alongshore sediment transport and spatially variable erosion.

Taken together, these results show that the hindcast performs reasonably well in identifying where storms come from, but is less reliable in representing how strong storms are and how long they persist. These biases and uncertainties propagate into storm classification and erosion analysis. Consequently, storm–shoreline relationships derived from the hindcast must be interpreted with caution, reinforcing the need for a probabilistic rather than deterministic interpretation of storm-driven coastal erosion in the following sections.

4.3 Shoreline response to storm type

To examine how different storm types affect the Canterbury coastline, monthly shoreline change derived from CoastSat was linked to storm types assigned through the hindcast–buoy storm matching. This allows observed erosion and accretion to be analysed as a function of hindcast-defined storm regime.

Table 2. Monthly shoreline change summary

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
mean_shoreline_change	313				0.065	1.4	-6.091	-0.711	0.064	0.868	5.757
ecan_storm_hours	300				23.92	13.43	1	14	22	31.25	75
ecan_max_Hs	300				2.941	0.277	2.56	2.79	2.87	3.013	5.47
ecan_mean_Hs	300				2.627	0.081	2.55	2.59	2.609	2.631	3.168
ecan_mean_dir	300				151.02	30.8	62.14	130.5	153.5	174.9	234.9
model_storm_hours	183				11.071	9.569	1	4	8	15	54
model_max_Hs	183				5.371	1.032	4.028	4.506	5.133	6.038	8.713
model_mean_Hs	183				4.665	0.443	4.028	4.301	4.606	4.899	6.45
model_mean_dir	183				182.8	34.82	45.783	180.6	197.4	202	224.1
erosion_month	314	2	FALSE	163							

Table 2 summarises the distribution of monthly shoreline change and storm forcing. The median shoreline change is close to zero (0.06 m), indicating that most months are approximately in equilibrium, but the interquartile range (−0.71 to 0.86 m) shows that changes of around a metre are common. Extreme months exhibit retreat greater than 6 m or accretion exceeding 5 m, highlighting the episodic, storm-driven nature of coastal change.

Storms may last anywhere from an hour or less to 70 hours or longer. The median storm length is approximately 22 hours. The average wave height during these storms is usually close to 2.6 meters, while the average highest wave is usually greater than 3 meters and up to 5 or more meters. The historical record of storm events has very similar wave heights, but a shorter duration than the average length of storms; this supports what had been determined regarding historical records having biases that indicate high waves and short durations. The overlap of erosive and non-erosive months across similar storm conditions shows that storm energy alone does not determine shoreline response.

The relationship between storm type and shoreline response is shown in Figure 7.

Figure 7. Monthly shoreline change grouped by hindcast-defined storm type.

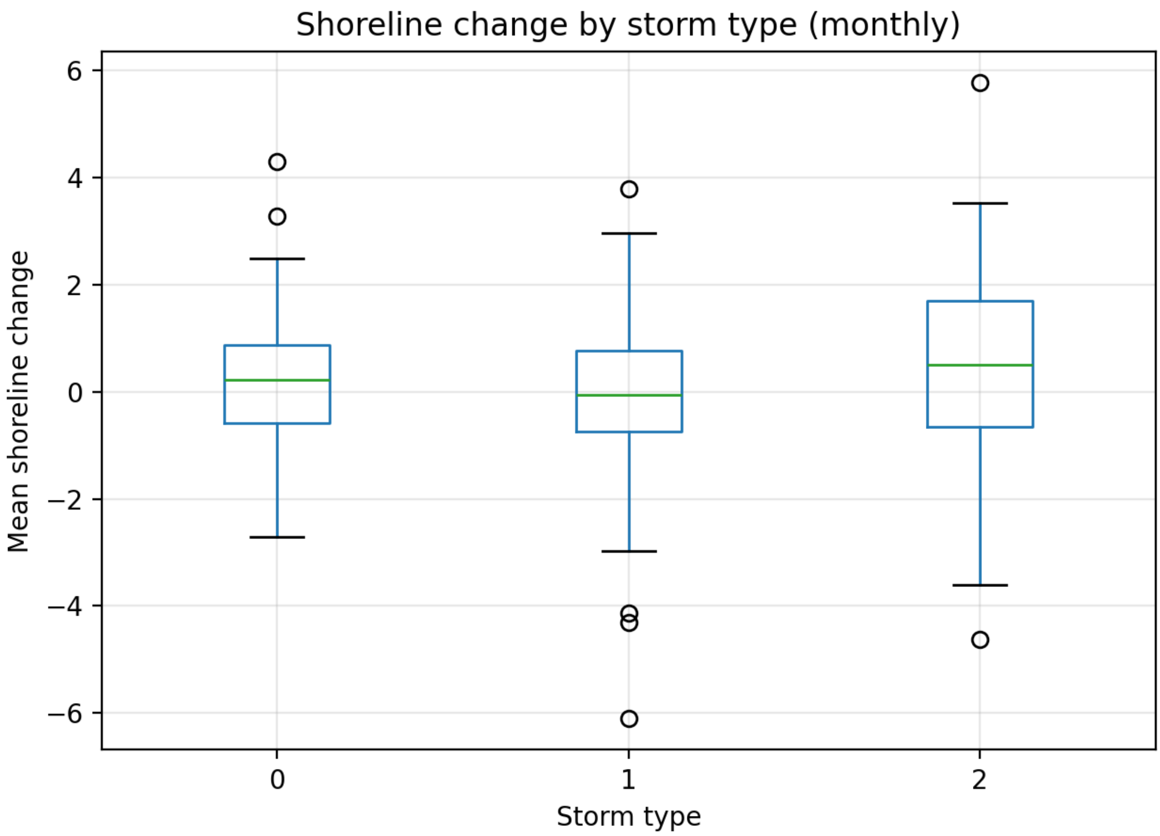


Figure 7 shows a clear dependence of shoreline change on storm type. Storm Type 1 has the most extreme negative outliers, with several months exhibiting shoreline retreat approaching 6 m. These long-duration, high-energy storms therefore dominate the most severe erosion events along the Canterbury coast. For Storm Type 1, the median shoreline movement is slightly negative which indicates that this storm type has an erosive tendency.

Storm Type 2 is the storm type with the most variability; it had the largest range of erosion and accretion and the highest median shoreline movement when compared to the previous storm type, with several points exceeding 5 m of shoreline advance. Storm Types 2 demonstrate that storms change rapidly and can cause erosion or recovery depending on how the coastline is oriented and if a beach is in poor condition prior to the storm.

Storm Type 0 is centred close to zero, with a narrower interquartile range than Storm Type 2 and fewer extreme outliers. These short, low-energy storms therefore represent relatively benign background conditions, during which the coastline remains largely stable or experiences modest recovery.

Both the pattern and statistics demonstrate that storm types separate low (Type 3) and high impact (Type 1) storm activity from those that are highly transient (Type 2), while also establishing an effective physical relationship between the characteristics of ocean waves generated by offshore winds and the reaction of the beach to storm activity.

4.4 Erosion prediction using storm characteristics

This section evaluates whether storm characteristics can be used to predict shoreline response. Two complementary approaches were used: **(i)** a regression-style assessment of how well storm variables explain monthly shoreline change magnitude, and **(ii)** a classification-style assessment of how well storm variables discriminate erosive months from non-erosive months.

Table 3. Model skill statistics for predicting monthly shoreline change.

MAE	RMSE	R^2	n_rows
1.040419	1.441802	0.05587	299

Table 3 indicates limited predictive skill for monthly shoreline change using storm variables alone. The model achieves **MAE = 1.04 m** and **RMSE = 1.44 m** across **n = 299** monthly observations, with a low explained variance (**R^2 = 0.056**). This implies that only about 6% of the month-to-month shoreline variability is captured by the available storm predictors, and that substantial shoreline change can occur even when storms appear similar in the offshore forcing data.

To interpret which storm variables contribute most strongly, logistic regression coefficients are summarised in Table 4. Logistic regression was selected because the outcome of interest is binary (erosion versus non-erosion), and the goal is to estimate erosion *risk* rather than exact shoreline change magnitude. Unlike black-box machine-learning methods, logistic regression allows direct interpretation of coefficients, making it possible to identify which storm characteristics increase or decrease erosion probability. This interpretability is particularly important for coastal management applications, where understanding drivers of risk is as important as prediction accuracy.

Table 4. Logistic regression coefficients used for erosion prediction.

feature	coefficient
ecan_max_Hs	-0.72046
dir_sin	-0.37023
type_0	-0.04854
type_2	-0.03307
type_1	0.062602
dir_cos	0.287358
ecan_mean_Hs	0.645871
ecan_storm_hours	0.912758

In Table 4, it is shown that both storm duration (ecan_storm_hours) and mean storm wave height (ecan_mean_Hs) have a strong positive effect on the fitted model; storm duration has a coefficient of 0.913, while mean significant wave height has a coefficient of 0.646. Although both directional terms have positive coefficients (dir_cos = 0.287) and negative coefficients (dir_sin = −0.370), the influence of direction approaches contributing to the storm events are less prominent than those of duration and intensity. Storm type variables have low coefficients (Type 1 = 0.063; Type 0 = −0.049; Type 2 = −0.033) suggesting that once we account for continuous storm metrics, storm types may provide little extra explanation.

To examine relationships among storm predictors and assess potential multicollinearity, a correlation matrix was computed between storm characteristics and monthly shoreline change (Figure 8).

Figure 8. Correlation Plot for storm characteristics vs Shoreline Response

Figure 8 shows strong correlations between several storm variables, particularly between maximum and mean wave height, and between hindcast-derived wave height metrics. These strong correlations explain why maximum wave height (ecan_max_Hs) appears with a negative coefficient in Table 5 despite being physically associated with energetic storms. The negative sign does not imply that large waves reduce erosion risk; rather, it reflects redistribution of explanatory power between correlated predictors in a multivariate model. This confirms that the regression coefficients should be interpreted cautiously and reinforces the probabilistic nature of the erosion prediction.

Model discrimination skill for erosion occurrence is summarised by the ROC curve.

Alternative approaches are commonly used in coastal erosion studies, including linear regression for shoreline change magnitude, decision trees and random forests for nonlinear classification, and Bayesian or process-based models that explicitly represent sediment transport. Previous studies have shown that while more complex models can improve predictive skill, they often sacrifice interpretability and require detailed nearshore data that are not available here. In this study, logistic regression was chosen as a baseline statistical model to test whether storm characteristics alone contain usable predictive signal, rather than to maximise predictive performance.

Figure 9. ROC curve for erosion prediction model.

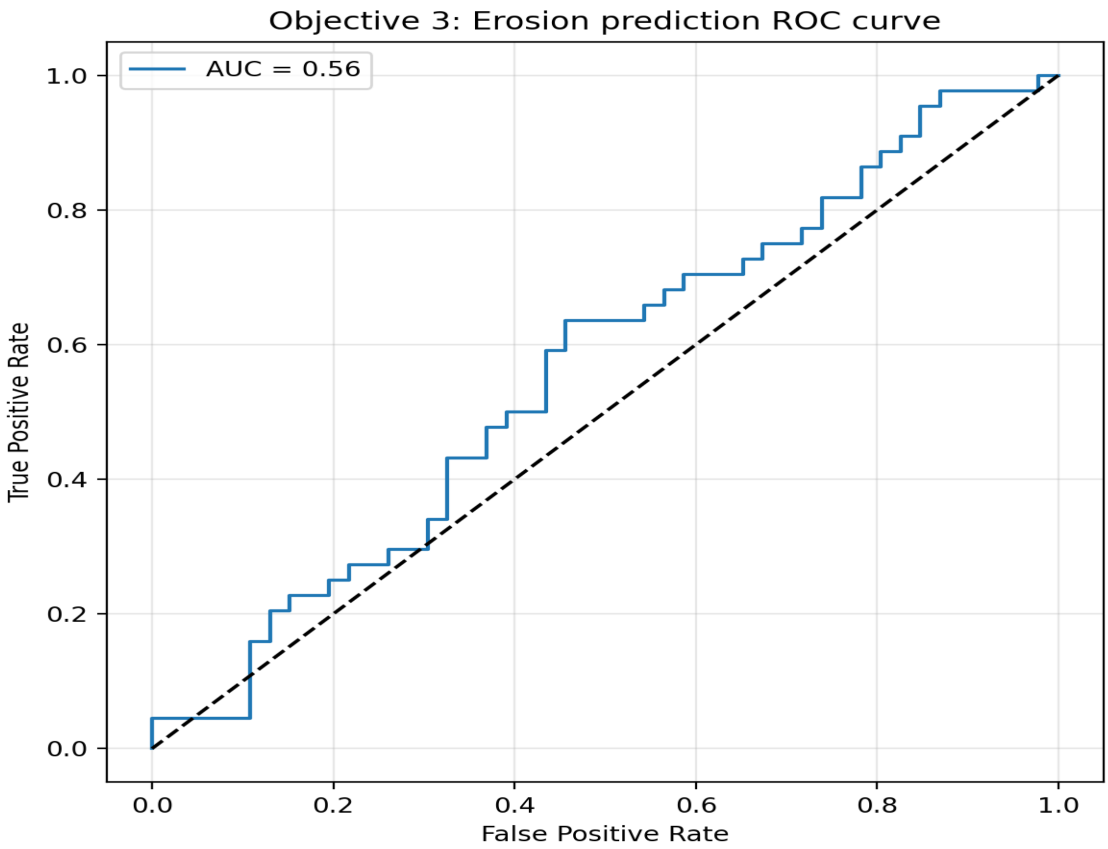


Figure 9 shows **AUC = 0.56**, only slightly above random classification (AUC = 0.50). This indicates that storm metrics contain some information about erosion risk but are insufficient, on their own, to reliably classify erosive months. Together with the low R^2 in Table 3, this confirms that shoreline response depends strongly on additional factors such as antecedent beach state, sediment supply, and storm sequencing. An AUC of 0.56 indicates performance only marginally better than random guessing, but still demonstrates that storm duration and intensity provide measurable, though limited, information about erosion likelihood.

4.5 Integrated storm–shoreline interpretation

The combination of storm type classification, historical validation of these classifications, observations of change to the shoreline, and predictions of future erosion provides an integrated assessment of the effects of storm-induced coastal change along the Canterbury coastline.

Storm typology results (Section 4.1; Figures 1–3) demonstrate that storms impacting the Canterbury region can be objectively grouped into three physically interpretable regimes based on wave height, duration, and approach direction. Storm Type 0 represents lower-energy, short-period storms that meet the storm threshold but typically produce relatively modest coastal forcing. Storm Type 1 consists of moderate-energy, mixed wave conditions and is the most common storm regime, representing background storm forcing. Storm Type 2 corresponds to high-energy swell-dominated storms, often associated with larger wave heights and longer durations, and therefore greater potential for coastal impact. This classification provides a simplified but physically meaningful description of the offshore storm climate affecting the region.

However, the hindcast–buoy validation (Table 1 and Figures 4–6) shows that the numerical wave hindcast used to define these storm types contains systematic biases. Significant wave heights are consistently overestimated, storm durations are compressed, and directional variability is underestimated, even though the dominant southerly storm direction is well captured. These limitations mean that hindcast-derived storm intensity does not translate directly into true nearshore forcing, and storm types should therefore be interpreted as relative regimes rather than exact physical magnitudes.

The results of the shoreline response analysis demonstrate a physical relationship between storm characteristics and the response of the beach to those storms, indicating that the storm type in general is correlated with how the beaches will respond (Table 2 & Figure 7). Storm patterns (Type 1) show the most significant amounts of shoreline erosion loss and the largest outliers for negative shore change and a bias towards erosion providing evidence to support this conclusion. Type 2 storms show the greatest variability among all storm types. There are times that these storms will lead to strong erosion and at other times lead to considerable accretion, which will depend upon the beachs' previous configuration and the coastal orientation. Type 0 storms generally reflect more stable conditions in terms of beach erosion and accretion due to being lower intensity storms and of shorter duration.

The erosion prediction results also clarify how limited the predictability of a storm can be in terms of predicting erosion events (Tables 3–5 & Figures 8-9). The duration of the storm event and the average wave height represent the strongest indicators of whether erosion will occur, while storm type indicators do not contribute anything to the predictability of erosion events once storm characteristics are included for analysis. The correlation analysis (Figure 8) demonstrates that there exists a strong interdependence among the wave height metrics, which also explains why there are coefficients in the regression equations that would appear to be counterintuitive. The Receiver Operating Characteristic Analysis (ROC) indicated a modest ability (AUC ≈ 0.56) to discriminate between storm types based on the storm characteristics alone and support the assertion that storm characteristics alone cannot provide enough information for accurate erosion predictions.

Taken together, these results demonstrate that storms are a necessary but not sufficient driver of shoreline change. Storm typology provides a useful framework for separating low-impact background conditions from erosion-prone storm regimes, but shoreline response remains highly variable. Variability is due to the influence of additional factors that are not considered in the offshore storm metrics. Additional factors can include previous beach states, availability of sand and sediment to be transported onto and off the beach, processes that happen nearshore, as well as various storm sequences.

Overall, the integrated results support a probabilistic interpretation of coastal erosion. Storm characteristics increase or decrease erosion risk, but similar storms can produce very different shoreline responses. This finding reinforces the importance of using storm-based analyses as a risk assessment tool rather than a deterministic predictor of coastal change.

5. Conclusion

This study investigated how storm characteristics influence shoreline change along the Canterbury coast by integrating wave hindcast data, ECan buoy observations, and satellite-derived shoreline positions. The results demonstrate that storm events can be grouped into physically meaningful types that differ in intensity, duration, and approach direction, and that these differences are reflected in observed shoreline response.

Hindcast storm clustering identified three dominant storm regimes. Weak, short-duration background storms (Storm Type 0) are associated with relatively stable or mildly accretionary shoreline behaviour. Storms with intermediate characteristics (Storm Type 1) show increased variability in shoreline response, while long-duration, high-energy storm regimes (Storm Type 2) produce the most severe erosion events, including the largest negative shoreline changes. This confirms that storm typology provides a useful framework for separating low-impact conditions from erosion-driving storm regimes.

Validation against buoy observations showed that the hindcast model captures dominant storm directions reasonably well but systematically misrepresents storm intensity and duration. These biases limit the accuracy of erosion prediction when relying on offshore wave data alone and highlight the importance of validation when hindcast products are used for coastal hazard assessment.

Erosion prediction using storm characteristics showed limited predictive skill. While storm metrics contain information about erosion likelihood, the results indicate that shoreline response is probabilistic rather than deterministic. Comparable storms do not always produce similar shoreline change, reflecting the influence of antecedent beach state, sediment availability, storm sequencing, and local coastal morphology.

The present findings support the body of research existing on the coastal erosion where storm length and wave energy are the main controls on extreme retreat of the shoreline in high energy environments. Additionally similar storm clustering and satellite-based shoreline data studies indicate significant variability in shoreline response under similar offshore forcing, indicating that the treatment of erosion as a risk-based process is much more beneficial than treating it as a deterministic event.

In conclusion, this research emphasizes the importance of using a typology of storms to better understand the potential for erosion from storm conditions; however, the predictive ability of storm conditions on future shoreline changes is limited. Future studies should focus on how to improve representation of nearshore processes, to include antecedent beach state, and to include accounts for storm sequences to improve assessments of coastal erosion risk in New Zealand’s high energy coastlines.

6. Future Work

The study illustrates that just determining offshore storms will provide an incomplete understanding of shoreline changes, and indicates a number of areas where further research could improve predictions of shoreline changes

First, hindcast wave models do not yet have enough accuracy for storm forcing, and there was consistent overestimation of wave height and underestimated duration of storms as well as loss of directional variability when validated with buoy data.. Future work should apply statistical bias correction or data assimilation techniques to align hindcast storm energy and persistence more closely with observed conditions before using these data in erosion models.

Second, coastal state variables should be incorporated into erosion prediction. The low skill of the statistical model indicates that antecedent beach width, dune condition, and sediment volume strongly influence how storms translate into erosion. Integrating CoastSat-derived beach width, pre-storm shoreline position, or intertidal slope into the modelling framework would likely improve predictive power and help explain why similar storms sometimes produce opposite coastal responses.

Third, storm sequencing should be considered. The impact of a storm depends not only on its own properties but also on the storms that preceded it. A moderate storm following a large erosive event may cause little additional retreat, whereas the same storm following a long calm period may produce substantial erosion. Future models could incorporate cumulative wave energy or time since the last major storm to better represent this dependency.

Finally, the framework could be extended spatially. The current analysis aggregates shoreline change across transects and months, but storm impacts vary along the coast depending on orientation, exposure, and sediment pathways. Applying the storm typology and modelling framework at the level of individual transects or embayments would allow more spatially resolved hazard assessments and support targeted coastal management.

7. Acknowledgements

This project made use of multiple external data sources and software tools. Wave buoy data were provided by Environment Canterbury (ECAN). Hindcast wave model data were obtained from regional numerical wave simulations for ECan buoy. Shoreline position data were derived using the CoastSat framework, which applies satellite imagery and computer vision techniques to extract shoreline locations.

This work was developed using Python and associated scientific libraries for data analysis, clustering, statistical modelling, and visualisation. Generative AI tools, including ChatGPT,

were used to assist with structuring, editing, and refining the written report, in accordance with DATA601 guidelines.

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9. Appendices

Appendix A – Data sources and preprocessing

Three principal datasets are compiled in this project: offshore wave observations from an ECan buoy; hindcast model results for waves in ECan buoy; and shoreline locations derived from satellites (CoastSat).

Wave heights offshore were obtained from the Environment Canterbury wave buoy positioned along the Canterbury coastline. This buoy produces a time series of Hs, wave periods and mean wave directions, with a high temporal resolution. These data represent the best available in-situ measurements of storm wave conditions and were used as the reference dataset for model validation and temporal alignment with shoreline change.

Hindcast wave data were obtained from a regional numerical wave model covering ECan buoy and the Canterbury shelf. The hindcast provides continuous time series of significant wave height, mean wave direction, and wave period at a fixed offshore grid point. These data were used to define storm events and storm typology because of their spatial and temporal consistency, which is not available from the buoy alone.

Shoreline position data were derived using the CoastSat framework, which applies automated shoreline detection to satellite imagery. Monthly median shoreline positions were extracted for a set of transects along the Canterbury coastline. Shoreline change was calculated as the difference between consecutive monthly median positions, with negative values indicating erosion and positive values indicating accretion.

All datasets were aligned to a common monthly time base. Storms detected in both hindcast and buoy datasets were associated with the month in which they occurred. When multiple storms occurred within a month, their metrics were aggregated to represent the overall storm forcing for that month. Shoreline change values were then matched to the corresponding storm forcing period to create the modelling dataset.

Appendix B – Storm detection and clustering

Independent identification of storm events was made through a combination of datasets reflecting buoys and hindcasts. A threshold-based method was employed to identify periods where the significant wave heights exceeded an arbitrary threshold for an arbitrarily predetermined minimum duration. Summary statistics were generated for each identified storm event to show peak significant wave height, mean significant wave height, and storm duration and mean wave direction.

Storm typology was developed from storm metrics derived from hindcasts. The identified hindcast storms were clustered based on peak significant wave height, mean significant wave height, storm duration, and mean wave direction. The parameters used to establish the clusters represent the relative magnitude, duration, and directional forces of each storm, which influence how the coastline will respond.

Three clusters were selected based on visual separation and physical interpretability. The resulting storm types correspond to: (i) short, low-energy storms; (ii) long-duration, high-energy storms; and (iii) short-duration but high-intensity storms approaching from multiple directions. These clusters form the storm typology used throughout the report.

To link hindcast storm types to observed coastal change, ECan buoy storms were matched to hindcast storms based on temporal overlap. Each buoy storm was assigned the storm type of its matched hindcast event. This allowed shoreline change, which is aligned with buoy timing, to be analysed in terms of hindcast-defined storm types.

Appendix C – Hindcast–buoy storm matching

Storm events were detected independently in the hindcast and ECan buoy datasets. To ensure that hindcast-defined storm types could be related to observed coastal response, individual buoy storms were matched to hindcast storms based on temporal overlap.

A buoy storm was considered to correspond to a hindcast storm if the time window of the buoy storm overlapped with that of a hindcast storm. When multiple hindcast storms overlapped a single buoy storm, the hindcast storm with the greatest temporal overlap was selected. Buoy storms with no overlapping hindcast storm were excluded from storm-type-based analysis.

Once matched, each buoy storm was assigned the storm type of its corresponding hindcast storm. This procedure ensured that storm types were defined in spatially consistent hindcast space while shoreline change and erosion were evaluated using locally observed buoy timing.

This storm matching process underpins all analyses that relate storm type to shoreline change, hindcast validation, and erosion prediction.

Appendix D – Erosion model specification

The logistic regression framework was used for erosion prediction. The response variable was binary and indicated the occurrence of net erosion (negative shoreline change) for a given month. The monthly shoreline changes used to determine net erosion were obtained from the CoastSat database as median monthly shorelines.

Predictors of net erosion included; the total duration of all storms within the month (storm duration); the average significant wave height for the month (mean significant wave height); the maximum significant wave height for the month (maximum significant wave height); and storm approach direction represented by sine and cosine transformations of the storm approach direction (storm approach direction).

Hindcast-defined storm type indicators were also included to test whether categorical storm regimes improved predictive performance beyond continuous storm metrics.
All predictors were used in a single multivariate model. The fitted coefficients are reported in Table 5. Model performance was assessed using standard regression and classification metrics, including mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and area under the ROC curve (AUC), as shown in Table 4 and Figure 7.

Both the extent and frequency of erosion can be assessed through this specification in a prediction based on storm activity and also provides an indication of limitations for predicting erosion from storms in a dynamic coastal environment.